

Expense Tracker using Python

Objective : The objective of this project is to develop a simple Python-based Expense Tracker application that helps users record daily expenses, categorize them, and view expense summaries. This project helps in understanding Python basics such as lists, dictionaries, loops, and conditional statements.

Problem Statement : Managing daily expenses manually can be difficult and error-prone. There is a need for a simple program that allows users to store expense details, organize expenses into categories, and view how much money is spent in each category. This project solves the problem by providing a menu-driven Python application to track expenses efficiently.

Features Implemented

- Menu-driven interface
- Add new expense categories
- Prevent duplicate category names
- Add expenses with date, amount, and category
- Validate expense amount input
- Validate category before adding expense
- View total expenses category-wise
- View all available categories
- Exit the program safely

Data Structures Used

- **List (*categories*)**
Used to store expense categories.
- **List of Dictionaries (*expenses*)**
Each expense is stored as a dictionary with keys:
 - *date*
 - *amount*
 - *category*
- **Dictionary (*expenses_summary*)**
Used to calculate and store total expenses for each category.

```

#Title : Expense Tracker
# List to store categories
categories = ['electronics','printers','entertainment']

# List to store expenses
expenses = []

while True: #To make infinite loop
    print('\n -----Expense Tracker-----')
    print('1.Add category')
    print('2.Add expense')
    print('3.view expenses')
    print('4.view categories')
    print('5.exit')

    choice = input('Enter your choice 1-5:') # to select any number from 1-5
    if choice == '1':
        print('Add category selected')
        new_category = input('Enter new category: ') # if we want to add new
category
        if new_category in categories:
            print('Category name already exists')
        else:
            print('category added successfully')
            categories.append(new_category) # after adding new category it will
append to categories

    elif choice == '2':
        print('Add expense selected')

        date = input('Enter date')
        amount = input("Enter amount")
        category = input('Enter category')
        print('Date entered:', date)
        print('amount entered:', amount)
        print('category entered:', category)
        try: # try block is used to check if the amount value is float or any
other datatype
            amount = float(amount)
            print('Amount valid')
        except ValueError:
            print('Amount not valid')
            continue
        if category not in categories:
            print('Category not valid')
            continue
        new_expense = {'date': date,
                        'amount' : amount,

```

```

        'category': category}
    expenses.append(new_expense)
    print('Expense added successfully')

elif choice == '3':
    print('View expenses selected')
    if expenses in []:
        print('No expenses')
    else:
        expenses_summary = {} # all the expenses are calculated in
expenses_summary

        for expense in expenses:
            category = expense['category']
            amount = expense['amount']

            if category in expenses_summary:
                expenses_summary[category] += amount
            else:
                expenses_summary[category] = amount

        print('\n expense summary:')
        for category, total in expenses_summary.items():
            print(category,':',total)

elif choice == '4':
    print('View categories selected')
    if categories in []:
        print('No categories')
    else:
        print('Available categories')
        for category in categories:
            print('-',category)

elif choice == '5':
    print('-----Exiting the program-----')
    break
else:
    print('-----Invalid choice-----')

```

Sample outputs

```
D:\SkillIT\Python\.venv\Scripts\python.exe D:\SkillIT\Python\Assignments\Expense_tracker.py
```

```
-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:4
View categories selected
Available categories
- electronics
- printers
- entertainment
```

```
-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:2
Add expense selected
Enter date25/03/2025
Enter amount1299.50
Enter categoryprinters
Date entered: 25/03/2025
amount entered: 1299.50
category entered: printers
Amount valid
Expense added successfully
```

```
-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:1
Add category selected
Enter new category: electronics
Category name already exists
```

```
-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:2
Add expense selected
Enter date23/01/2025
Enter amount800
Enter categoryelectronics
Date entered: 23/01/2025
amount entered: 800
category entered: electronics
Amount valid
Expense added successfully
```

-----Expense Tracker-----

- 1.Add category
- 2.Add expense
- 3.view expenses
- 4.view categories
- 5.exit

Enter your choice 1-5:2

Add expense selected

Enter date14/04/2024

Enter amount335.46

Enter categoryentertainment

Date entered: 14/04/2024

amount entered: 335.46

category entered: entertainment

Amount valid

Expense added successfully

-----Expense Tracker-----

- 1.Add category
- 2.Add expense
- 3.view expenses
- 4.view categories
- 5.exit

Enter your choice 1-5:3

View expenses selected

expense summary:

electronics : 800.0

printers : 1299.5

entertainment : 335.46

-----Expense Tracker-----

- 1.Add category
- 2.Add expense
- 3.view expenses
- 4.view categories
- 5.exit

Enter your choice 1-5:2

Add expense selected

Enter date03/06/2025

Enter amount512.86

Enter categoryelectronics

Date entered: 03/06/2025

amount entered: 512.86

category entered: electronics

Amount valid

Expense added successfully

-----Expense Tracker-----

- 1.Add category
- 2.Add expense
- 3.view expenses
- 4.view categories
- 5.exit

Enter your choice 1-5:3

View expenses selected

-----Expense Tracker-----

- 1.Add category
- 2.Add expense
- 3.view expenses
- 4.view categories
- 5.exit

Enter your choice 1-5:4

View categories selected

Available categories

- electronics
- printers
- entertainment
- gaming

```
-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:9
-----Invalid choice-----

-----Expense Tracker-----
1.Add category
2.Add expense
3.view expenses
4.view categories
5.exit
Enter your choice 1-5:5
-----Exiting the program-----
Process finished with exit code 0
|
```

Conclusion : The Expense Tracker application was successfully developed using Python. The program allows users to manage expenses efficiently with proper validation and clear output. This project strengthened understanding of Python fundamentals such as loops, conditionals, lists, dictionaries, and error handling.

Future Enhancements

- Add date format validation
- Allow deletion or modification of expenses